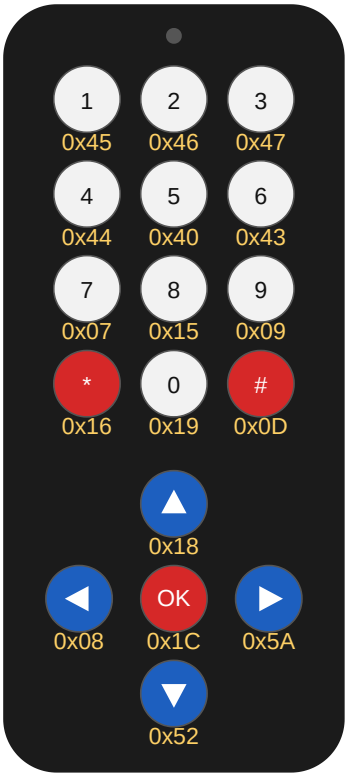
NEC IR remote control HX1838 – code table

17-key remote from Arduino kits · address 0x00 · 38 kHz carrier



Command code (hex) below each key

Frame format

Leader 9 ms burst + 4.5 ms space

Byte 1 Address = **0x00**

Byte 2 ~Address = **0xFF**

Byte 3 Command (see table)

Byte 4 ~Command (check)

Stop 562.5 µs burst

Each byte is sent **LSB first**. Byte 3 + byte 4 = 0xFF in a valid frame.

Bit encoding

Logic 0 562.5 µs burst + 562.5 µs space = 1.125 ms

Logic 1 562.5 µs burst + 1687.5 µs space = 2.25 ms

Burst ≈ 21 periods of 38 kHz (26.3 µs)

Repeat code

While the key is held, the remote only sends
9 ms burst + 2.25 ms space + 562.5 µs burst
about every 108 ms (measured from the start of the frame).

Receiver (HX1838 / TSOP)

The output is **active low**: low during a burst, high during space and idle.
Decode by measuring the **high time** after each low pulse.

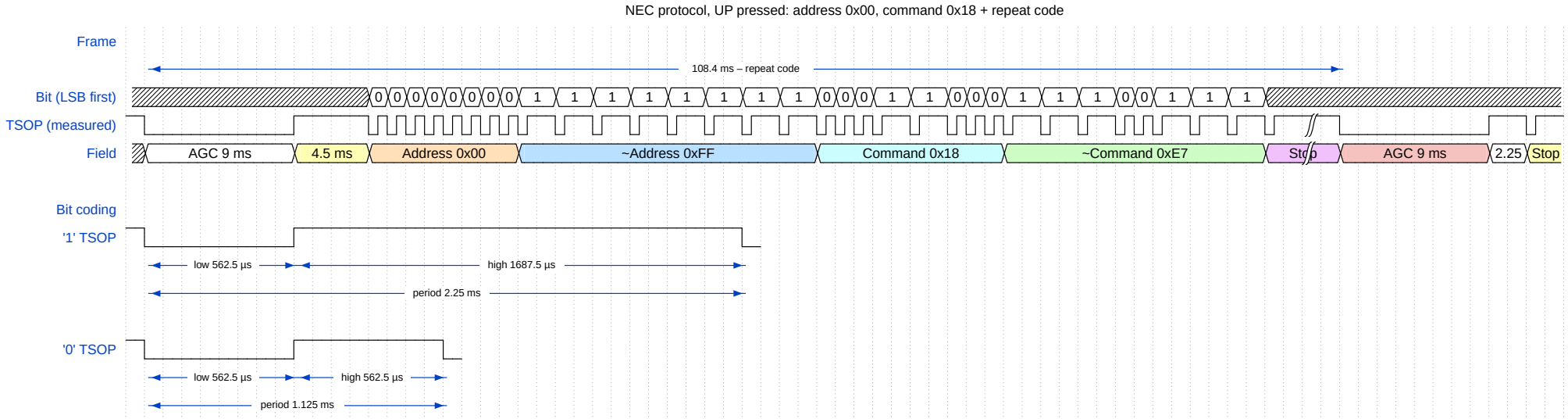
Code table

Key	Command	~Command	Command bits LSB first, as transmitted	IRremote 2.x results.value	IRremote 4.x decodedRawData
1	0x45	0xBA	10100010	0x00FFA25D	0xBA45FF00
2	0x46	0xB9	01100010	0x00FF629D	0xB946FF00
3	0x47	0xB8	11100010	0x00FFE21D	0xB847FF00
4	0x44	0xBB	00100010	0x00FF22DD	0xBB44FF00
5	0x40	0xBF	00000010	0x00FF02FD	0xBF40FF00
6	0x43	0xBC	11000010	0x00FFC23D	0xBC43FF00
7	0x07	0xF8	11100000	0x00FFE01F	0xF807FF00
8	0x15	0xEA	10101000	0x00FFA857	0xEA15FF00
9	0x09	0xF6	10010000	0x00FF906F	0xF609FF00
*	0x16	0xE9	01101000	0x00FF6897	0xE916FF00
0	0x19	0xE6	10011000	0x00FF9867	0xE619FF00
#	0x0D	0xF2	10110000	0x00FFB04F	0xF20DFF00
UP	0x18	0xE7	00011000	0x00FF18E7	0xE718FF00
LEFT	0x08	0xF7	00010000	0x00FF10EF	0xF708FF00
OK	0x1C	0xE3	00111000	0x00FF38C7	0xE31CFF00
RIGHT	0x5A	0xA5	01011010	0x00FF5AA5	0xA55AFF00
DOWN	0x52	0xAD	01001010	0x00FF4AB5	0xAD52FF00

UP (0x18) is verified against a measured signal (page 2). The other codes are the standard values for this type of remote – clones may differ, so check with your own receiver.

Measured signal: the UP key

nec_Bt_up_rec.csv · TSOP output sampled every 25 µs · address 0x00, command 0x18 + repeat code



Frame: 1 tick = 1.125 ms · Bit coding: to scale, 562.5 µs = 8 ticks · TSOP output is active low · gap (|) = 39 ms without signal, shortened

How to decode the signal

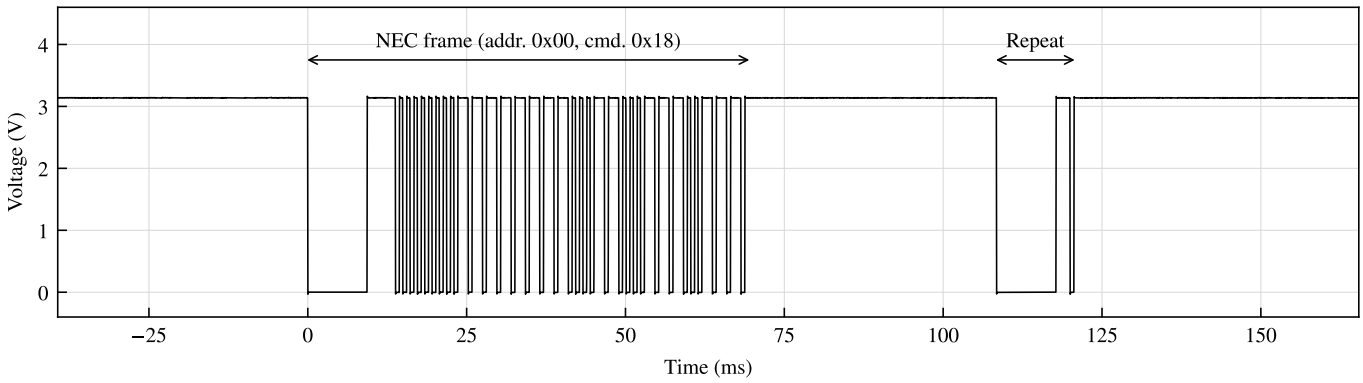
1. Wait for a falling edge (start of a burst).
2. Measure the low time ≈ 9 ms, then the high time: ≈ 4.5 ms = new frame, ≈ 2.25 ms = repeat code.
3. For each of the 32 bits: measure the high time after the short low pulse. ≈ 0.56 ms → 0, ≈ 1.69 ms → 1. Use a threshold of e.g. 1.1 ms.
4. Assemble the bytes with the first bit as bit 0 (LSB first).
5. Check that command + ~command = 0xFF.

Measured vs. nominal timing

	Nominal	Measured
Leader low	9.0 ms	9.35 ms
Leader high	4.5 ms	4.48 ms
Bit low (burst)	562.5 µs	600–625 µs
Bit high, 0	562.5 µs	525–550 µs
Bit high, 1	1687.5 µs	1625–1650 µs
Frame → repeat	≈ 108 ms	108.4 ms

The TSOP receiver stretches the low pulses slightly and shortens the spaces by the same amount. A decoder must therefore use tolerances, not exact times.

Oscilloscope measurement



Measured output of the TSOP IR receiver (active low) for the UP key: NEC frame with address 0x00 and command 0x18, followed by a repeat code 108.4 ms after the start of the frame.